Active Directory Infrastructure Troubleshooting Report

Sample Work — Demonstration Environment

Project Overview

This report documents Active Directory infrastructure troubleshooting performed in an AWS-hosted Windows Server environment.

The environment was configured to provide centralized identity and authentication services for Windows-based infrastructure and application delivery. Troubleshooting involved Active Directory, DNS, Windows Server, Amazon FSx for Windows File Server, and application delivery components.

The work demonstrates a structured approach to diagnosing infrastructure issues by examining connectivity, domain services, DNS, authentication, server configuration, and supporting infrastructure before making changes.

Environment

The demonstration environment included:

- AWS infrastructure
- Windows Server 2022
- Active Directory Domain Services
- DNS
- Amazon FSx for Windows File Server
- Parallels RAS application delivery
- Windows-based virtual desktop environment
- MSIX/VHDX application packages

Active Directory served as the centralized identity and authentication platform for the Windows infrastructure.

Challenge

The environment required reliable communication between Windows Server systems, Active Directory, DNS, shared storage, and application delivery services.

The troubleshooting process focused on verifying that:

- Domain services were available.
- Windows Server systems could communicate with the domain.
- DNS was resolving domain resources correctly
- Domain membership and authentication were functioning.
- Active Directory resources could be accessed as expected.
- Shared infrastructure could communicate with domain services.
- Application delivery components could operate within the domain environment.

Because multiple infrastructure components were interconnected, troubleshooting needed to determine whether an issue originated within Active Directory itself or from another supporting service.

Troubleshooting Approach

The troubleshooting process followed a layered approach:

1. Verify Domain Services

Checked Active Directory services and domain controller availability to confirm the domain was operational.

2. Verify DNS

DNS resolution was examined because Active Directory relies heavily on DNS for locating domain controllers and other domain services.

3. Verify Domain Connectivity

We checked Windows Server connectivity to the Active Directory environment to identify potential communication or configuration problems.

4. Verify Domain Configuration

Server domain membership, authentication behavior, and Active Directory configuration were reviewed.

5. Review System Evidence

Windows tools such as Event Viewer and PowerShell were used to examine system information, services, and errors that could identify the source of an issue.

6. Verify Supporting Infrastructure

Active Directory integration with supporting services, including Amazon FSx and application delivery components, was reviewed to determine whether the issue extended beyond the domain environment.

Tools Used

The troubleshooting process used standard Windows administration and diagnostic tools, including:

- Server Manager
- Active Directory Users and Computers
- DNS Manager
- Group Policy Management
- Event Viewer
- PowerShell
- Windows Explorer
- Windows Server administrative tools
- Parallels RAS management tools

PowerShell was used to retrieve Active Directory and domain information and to verify infrastructure configuration.

Active Directory Administration

The environment included standard Active Directory administration tasks involving:

- User accounts
- Computer accounts
- Organizational structure
- Domain configuration
- Domain controllers
- Authentication
- Group Policy
- DNS integration

The environment was configured to provide centralized management of Windows resources through Active Directory.

Application Delivery Integration

Active Directory was also used as part of the broader application delivery environment.

Windows Server, Active Directory, Amazon FSx, and Parallels RAS worked together to provide access to shared resources and published applications.

This required troubleshooting beyond Active Directory alone, including verifying domain communication and access to infrastructure resources used by the application delivery environment.

Findings

The troubleshooting process demonstrated the importance of separating Active Directory problems from issues occurring in connected infrastructure.

Several infrastructure components were functioning correctly, including Windows Server, Active Directory, DNS, and shared storage. This allowed troubleshooting to move beyond basic domain connectivity and focus on the specific component responsible for an application delivery issue.

The investigation also demonstrated that an apparent application problem can originate from dependencies or configuration outside of Active Directory itself.

Resolution

The environment was brought to an operational state through verification and correction of the underlying infrastructure configuration.

The troubleshooting process established:

- Active Directory was operational.
- Domain services were available.
- DNS supported the domain environment
- Windows Server systems could communicate with the domain.
- Shared infrastructure could be integrated with the Windows environment.
- Application delivery components could use the established infrastructure.

Issues that were found outside of Active Directory were isolated rather than incorrectly attributed to the domain.

Result

The completed troubleshooting process provided a functioning Windows infrastructure environment with Active Directory serving as the centralized identity and authentication platform.

The investigation also established a repeatable troubleshooting methodology for environments where Active Directory interacts with DNS, Windows Server, file storage, and application delivery platforms.

Key Takeaway

Effective Active Directory troubleshooting requires looking beyond the domain itself.

A structured investigation can move from:

Domain Services → DNS → Connectivity → Authentication → Configuration → Event Logs → Supporting Infrastructure

This approach helps identify the actual source of an issue and avoids unnecessary configuration changes.

Demonstration Environment Notice

This report describes work performed in a demonstration/lab environment and is provided as a sample of technical troubleshooting methodology and infrastructure experience.